We have, $x(m-n)+y(n-l)+z(l-m)$

$= a[(m-n)+(n-l)+(l-m)]+d[(m-n)(l-1)+(n-l)(m-1)+(l-m)(n-1)]$

$= a[0]+d[lm-ln-m+n+mn-lm-n+l+ln-mn-l+m]$

$= a(0)+d(0)=0$

(ii) On subtracting equation (2) from equation (1), equation (3) from equation (2) and equation (1) from equation (3), we get

$$x-y=(l-m)d$$

$$y-z=(m-n)d$$

$$z-x=(n-l)d$$

$$(x-y)n+(y-z)l+(z-x)m = [(l-m)n+(m-n)l+(n-l)m]d$$

$$= \left[ln-mn+lm-nl+nm-lm\right]d = 0$$

Note

In an Arithmetic Progression

- If every term is added or subtracted by a constant, then the resulting sequence is also an A.P.
- If every term is multiplied or divided by a non-zero number, then the resulting sequence is also an A.P.
- If the sum of three consecutive terms of an A.P. is given, then they can be taken as $a-d$, a and $a+d$. Here the common difference is d.
- If the sum of four consecutive terms of an A.P. is given then, they can be taken as $a-3d$, $a-d$, $a+d$ and $a+3d$. Here common difference is $2d$.

Example 2.29 In an A.P., sum of four consecutive terms is 28 and the sum of their squares is 276. Find the four numbers.

Solution Let us take the four terms in the form $(a-3d), (a-d), (a+d)$ and $(a+3d)$.

Since, sum of the four terms is 28,

$$a-3d+a-d+a+d+a+3d=28$$

$$4a=28 \Rightarrow a=7$$

Similarly, since sum of their squares is 276,

$$(a-3d)^2+(a-d)^2+(a+d)^2+(a+3d)^2=276.$$

$$a^2-6ad+9d^2+a^2-2ad+d^2+a^2+2ad+d^2+a^2+6ad+9d^2=276$$

$$4a^2+20d^2=276 \Rightarrow 4(7)^2+20d^2=276.$$

$$d^2=4 \Rightarrow d=\pm\sqrt{4} \text{ then, } d=\pm 2$$

If $d=2$ then the four numbers are $7-3(2)$, $7-2$, $7+2$, $7+3(2)$

That is the four numbers are 1, 5, 9 and 13.

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